1 Inputs and ROI

ERP panorama I



Instruction t

Reference-ROI (Grounded-SAM)

→ b, optional m_{erp}

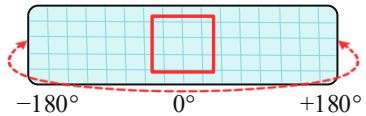
Predicted ROI (Grounding DINO)

↓

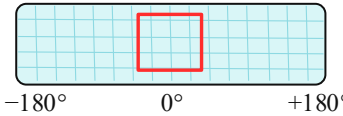
b, c $c \geq \tau$

Manual → λ_0, φhint$_0$, θ

2 Circular ROI canonicalisation



Shortest circular arc



Compute circular shift

$r = \text{round}_{hu}(W/2 - c(m_{circ})) \bmod W$

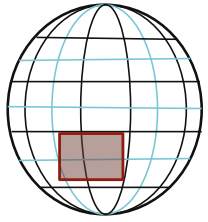
Apply joint shift → centred ROI

$(I_r, m_r) = (T_r I, T_r\, m_{circ})$

Set view and FoV

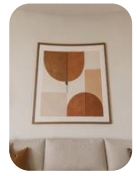
$V = (\lambda_0, \varphi_0, \theta x),\ 30° \leq \theta_x \leq 150°$

3 Gnomonic projection and mask



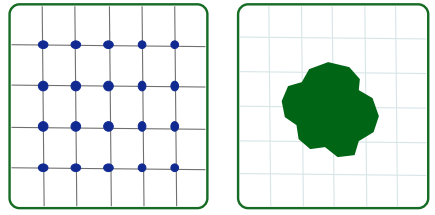
Periodic bilinear sampling

P



$P = \Pi(I_r; \lambda_0, \varphi_0, \theta), 512 \times 512$

S: patch → ERP map m_p: Projected Support



4 Frozen editor and patch constraint

P m_p $Q(P, m_p)$



$t' = t$ + optional AESG suffix

Frozen E (Qwen-Image-Edit-2511)

$P_E = E([P, Q(P, m_p)], t')$



$A_p = \mathcal{G}(m_p; \rho_p = 3\text{ px})$

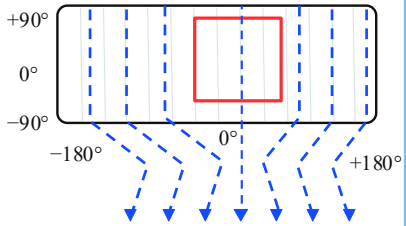


$P' = A_p \odot P_E + (1 - A_p) \odot P$



5 Dense analytic inverse warp

Dense ERP grid queries P'



$R = \mathcal{R}(P'; \lambda_0, \varphi_0, \theta)$

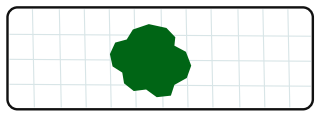
R



m_p + eroded frustum → inverse-warped Ω

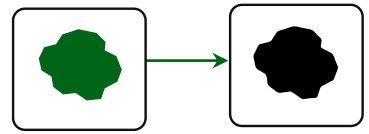
No forward-scatter holes

Ω



6 Feather, composite, and restore

Ω M = Feather(Ω), 8 px



$I^*_r = M \odot R + (1 - M) \odot I_r$

↓

$I^* = T_{-r}(I^*_r)$

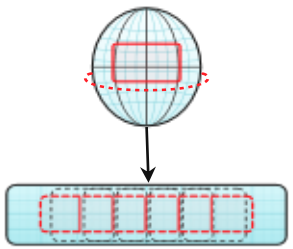
↓

EditEdited ERP I*



Outside Ω: unchanged

A. Wide-ROI tiling



Wrapped ROI: canonicalise first

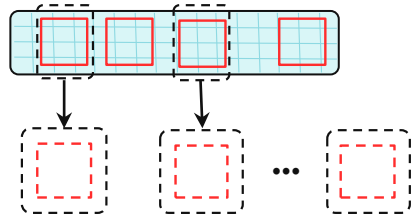
60° tiles · 20% overlap · $K \leq 6$

$I^{(k)} = \mathcal{A}(I^{(k-1)}, t, b_k)$

$\mathcal{L}$: single-ROI core

view + Stages 3–6

B. Repeated-target editing



Group under 110° FoV · $1 \leq G \leq 6$

$I^{(k)} = \mathcal{A}(I^{(k-1)}, t, b_k; m_k)$